

$$= \begin{vmatrix} 2 & 3 & 6 \\ 3 & 2 & 1 \\ 4 & 1 & 7 \end{vmatrix} + \begin{vmatrix} 6 & 3 & 6 \\ 4 & 2 & 1 \\ 2 & 1 & 7 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 3 & 6 \\ 3 & 2 & 1 \\ 4 & 1 & 7 \end{vmatrix} + 2 \begin{vmatrix} 3 & 3 & 6 \\ 2 & 2 & 1 \\ 1 & 2 & 7 \end{vmatrix}$$

$$= \Delta_1 + \Delta_2 \text{ (say)}$$

$$\therefore \Delta_1 = 2(14-1) - 3(21-4) + 6(3-8) = 26 - 51 - 30 = -55$$

$$\Delta_2 = 2[3(14-1) - 3(14-1) + 6(2-2)] = -55$$

$$\Delta = 8(14-1) - 3(49-6) + 6(7-12) = 104 - 129 - 30 = -55$$

$$\therefore \Delta = \Delta_1 = \Delta_2; \quad \text{Hence, the proposition is proved.}$$

5.2.4 Product of Two Determinants

The product of two determinants of same order is another determinant of the same order.

Example: $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} c_1 & d_1 \\ c_2 & d_2 \end{vmatrix} = \begin{vmatrix} a_1c_1 + b_1d_1 & a_1c_2 + b_1d_2 \\ a_2c_1 + b_2d_1 & a_2c_2 + b_2d_2 \end{vmatrix}$

Example: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} =$

$$\begin{vmatrix} a_1x_1 + b_1y_1 + c_1z_1 & a_1x_2 + b_1y_2 + c_1z_2 & a_1x_3 + b_1y_3 + c_1z_3 \\ a_2x_1 + b_2y_1 + c_2z_1 & a_2x_2 + b_2y_2 + c_2z_2 & a_2x_3 + b_2y_3 + c_2z_3 \\ a_3x_1 + b_3y_1 + c_3z_1 & a_3x_2 + b_3y_2 + c_3z_2 & a_3x_3 + b_3y_3 + c_3z_3 \end{vmatrix}$$

Numerically, $\begin{vmatrix} 4 & 5 & 9 \\ 1 & 2 & 3 \\ 3 & 0 & 1 \end{vmatrix} \times \begin{vmatrix} 2 & 1 & 5 \\ 3 & 2 & 1 \\ 1 & 6 & 4 \end{vmatrix}$

$$= \begin{vmatrix} 4x_2 + 5x_3 + 9x_1 & 4x_1 + 5x_2 + 9x_6 & 4x_5 + 5x_1 + 9x_4 \\ 1x_2 + 2x_3 + 3x_1 & 1x_1 + 2x_2 + 3x_6 & 1x_5 + 2x_1 + 3x_4 \\ 3x_2 + 0x_3 + 2x_1 & 3x_1 + 0x_2 + 2x_6 & 3x_5 + 0x_1 + 2x_4 \end{vmatrix} = \begin{vmatrix} 32 & 68 & 61 \\ 11 & 23 & 19 \\ 8 & 15 & 23 \end{vmatrix}$$

Check Your Progress 1

1) Find the value of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

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2) You are given a determinant $\begin{vmatrix} 2 & 0 & -1 \\ 1 & 1 & 7 \\ 3 & 3 & 9 \end{vmatrix}$. Work on it to verify important properties.

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3) Evaluate $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

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4) Find the value of $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$

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5.2.5 Adjoint and Reciprocal Determinants

The ‘*adjoint*’ or ‘*adjugate*’ of a determinant Δ is the determinant Δ' whose elements are the cofactors of the corresponding elements of Δ

$$\text{Let } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ then, } \Delta' = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$$

Where $A_1, B_1, C_1, \dots$ are the respective cofactors of $a_1, b_1, c_1, \dots$ of Δ .

Note: Here $\Delta' = \Delta^2$ when $\Delta \neq 0$.

Example: Let $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix}$

Here, $a_1 = 0, b_1 = 1, c_1 = 2$

$$a_2 = 2, b_2 = 0, c_2 = 1$$

$$a_3 = 1, b_3 = 2, c_3 = 0$$

$$\therefore \Delta_1 = \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = 0 - 2 = -2$$

$$B_1 = (-1) \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -(0 - 1) = +1$$

Similarly, $C_1 = 4, A_2 = 4, B_2 = -2, C_2 = 1,$

$$A_3 = 1, B_3 = 4, C_3 = -2$$

$$\therefore \text{Adjoint of } \Delta \text{ is } \Delta' = \begin{vmatrix} -2 & 1 & 4 \\ 4 & -2 & 1 \\ 1 & 4 & -2 \end{vmatrix}$$

The ‘*reciprocal*’ or ‘*inverse*’ of a determinant $\Delta (\neq 0)$ is the determinant, which is formed by dividing every element of the adjoint of Δ by Δ . Thus, the reciprocal of Δ is

$$\begin{vmatrix} \frac{A_1}{\Delta} & \frac{B_1}{\Delta} & \frac{C_1}{\Delta} \\ \frac{A_2}{\Delta} & \frac{B_2}{\Delta} & \frac{C_2}{\Delta} \\ \frac{A_3}{\Delta} & \frac{B_3}{\Delta} & \frac{C_3}{\Delta} \end{vmatrix} = \frac{1}{\Delta^3} \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = \frac{1}{\Delta^3} \cdot \Delta' = \frac{1}{\Delta^3} \cdot \Delta^2 = \Delta^{-1}$$

Example: Following the last example (i.e., Example 11), we get,

$$\Delta = 0(0 - 2) + 1(1 - 0) + 2(4 - 0) = 9 \neq 0$$

$$\therefore \text{The reciprocal of } \Delta \text{ is } \begin{vmatrix} -\frac{2}{9} & \frac{1}{9} & \frac{4}{9} \\ 4/9 & -2/9 & 1/9 \\ 1/9 & 4/9 & -2/9 \end{vmatrix}$$

5.2.6 Symmetric Determinants

If a 3 x 3 determinant is written in the form $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

in which the suffix of any element indicates its position (e.g., the suffix 12 of the element a_{12} indicates that the element is in the 1st row and in the 2nd column) and if $a_{ij} = a_{ji}$ for all $i, j = 1, 2, 3$; then the **determinant is said to be symmetric**.

Thus $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ is a symmetric determinant.

The above definition of symmetric determinant can be extended similarly to determinants of order n.

Properties: <a>The square of any symmetric determinant is a symmetric determinant.

The adjoint of symmetric determinant is symmetric.

Example: Let $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix}$

Here Δ is a symmetric determinant

$$\text{Now, } \Delta^2 = \Delta \cdot \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix} \times \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix}$$

$$= \begin{vmatrix} 1+4+9 & 2+8+15 & 3+10+21 \\ 2+8+15 & 4+16+25 & 6+20+35 \\ 3+10+21 & 6+20+35 & 9+25+49 \end{vmatrix}$$

$$= \begin{vmatrix} 14 & 25 & 34 \\ 25 & 45 & 61 \\ 34 & 61 & 83 \end{vmatrix}, \text{ which is also a symmetric determinant.}$$

$$\text{Now adj. } \Delta = \begin{vmatrix} 3 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 0 \end{vmatrix}, \text{ which is also a symmetric determinant}$$

5.2.7 Skew and Skew-Symmetric Determinants

The determinant
$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

is said to be 'skew', if $a_{ij} = -a_{ji} \forall i, j = 1, 2, 3$ and $i \neq j$

Thus,
$$\begin{vmatrix} a & -b & -g \\ h & b & -f \\ g & f & c \end{vmatrix}$$
 is a **skew** determinant.

And if $a_{ij} = -a_{ji} \forall i, j = 1, 2, 3$ and $i \neq j$ with $a_{ii} = 0 \forall i = j$,

then the determinant is said to be skew-symmetric.

Thus,
$$\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$$
 is a **skew-symmetric** determinant.

Properties: <a>Every skew-symmetric determinant of third order (more generally, of odd order) is zero.,

Every skew-symmetric determinant of second order (more generally of even order) is a perfect square.

Example: Let
$$\Delta = \begin{vmatrix} 0 & 5 & 2 \\ -5 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = (-1)^3 \begin{vmatrix} 0 & -5 & -2 \\ 5 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix}$$

[taking (-1) common from each column]

$$= -\Delta$$

Hence, $2\Delta = 0 \Rightarrow \Delta = 0$

Again, let us consider an even order determinant

$$\Delta_1 = \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix}$$

$$\therefore \Delta_1 = 0 - (-4) = 4 = 2^2$$
 which is a perfect square.

5.2.8 Solution of Simultaneous Equations by Cramer's Rule

Gabriel Cramer obtained a simple method of solution of linear simultaneous equations by using the properties of determinants.

In particular, let the system of 3 equations in 3 variables, be

$$a_i x + b_i y + c_i z = k_i \quad (i = 1, 2, 3) \dots\dots\dots (1)$$

$$\text{and } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$$

$$\text{Then } x\Delta = \begin{vmatrix} xa_1 & b_1 & c_1 \\ xa_2 & b_2 & c_2 \\ xa_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix} [c_i' = c_1 + yc_2 + zc_3]$$

$$= \begin{vmatrix} k_1 & b_1 & c_1 \\ k_2 & b_2 & c_2 \\ k_3 & b_3 & c_3 \end{vmatrix} \text{ [from (1)]} = \Delta_1 \text{ (say)}$$

$$\therefore x = \frac{\Delta_1}{\Delta}$$

Similarly, $y = \frac{\Delta_2}{\Delta}$ and $z = \frac{\Delta_3}{\Delta}$ where Δ_2 and Δ_3 are the determinants obtained from Δ substituting the columns (k_1, k_2, k_3) for the second and third column respectively.

Example: Suppose that the numbers x_1 and x_2 satisfy the equations

$$x_1 - 2x_2 = 3 \text{ and } 3x_1 + 5x_2 = 20.$$

Find x_1 and x_2 by using Cramer's rule.

$$\text{Ans. } x_1 - 2x_2 = 3, 3x_1 + 5x_2 = 20$$

It may be written as matrix equation $AX = B$

$$\text{where, } A = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}, AX = B \text{ } X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 \\ 20 \end{pmatrix}$$

Cramer's rule tells us that

$$x_1 = \frac{\begin{vmatrix} 3 & -2 \\ 20 & 5 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix}} = \frac{3(5) - (-2)(20)}{1(5) - (-2)3} = \frac{55}{11} = 5$$

$$\text{and } x_2 = \frac{\begin{vmatrix} 1 & 3 \\ 3 & 20 \end{vmatrix}}{\begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix}} = \frac{1(20) - 3(3)}{11} = \frac{11}{11} = 1$$

Thus, if we have the value of the original determinant along with the values obtained from substituting the column of constants (i.e., k_i) in different places, then by Cramer's rule we can have the value of the corresponding variable.

Check Your Progress 2

- 1) Determine the signs to be attached to relevant minors if you are given the co-factors of a determinant as $|C_{13}|$, $|C_{23}|$, $|C_{33}|$, $|C_{41}|$ and $|C_{34}|$.

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- 2) Solve by Cramer's rule $x + 2y + 3z = 6$

$$2x + 4y + z = 7$$

$$3x + 2y + 9z = 14$$

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5.3 MATRIX

5.3.1 Concept of Matrix

In real life, we do not have to deal with a single magnitude but with a set of magnitudes simultaneously. In analysis we treat such a set of magnitudes as a single entity or a single object of thought, abstracted by numbers. Let us consider a problem where we treat a set as single entity and a number of such sets as a single system. We will then examine how the fundamental operations of arithmetic can be suitably defined to deal with such a system.

Suppose there are 3 individuals A, B, C in a family and that

A has a set of 3 pants, 3 shirts and 1 tie.

B has a set of 5 pants, 5 shirts and 2 ties.

C has a set of 6 pants, 8 shirts and no tie.

We can arrange the data in the following convenient system:

<i>Individuals</i>		<i>Pants</i>	<i>Shirts</i>	<i>Ties</i>		
	↓	(P)	(S)	(T)		
A	→	3	3	1	→	1 st Row
B	→	5	5	2	→	2 nd Row
C	→	6	8	0	→	3 rd Row
		↓	↓	↓		
		1 st	2 nd	3 rd		
		Col.	Col.	Col.		

This system comprises 3 rows and 3 columns (note that, by convention, rows are written first and columns afterwards). We have written belongings of A in the first row, belongings of B in the second row and the belongings of C in the third row. Clearly, 1st column gives us the numbers of pants that A, B and C respectively have, while second column gives us shirts of A, B, and C respectively; and similarly, third column gives the number of ties of A, B, C respectively.

The arrangement of numbers written in such a particular form of rows and columns and enclosed by [], the square brackets, or (), the large parentheses, is called a '**matrix**'. In general,

A '**matrix**', in general form, is defined as a rectangular array of elements arranged in rows and columns as shown below

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots a_{1n} \\ a_{21} & a_{22} & \dots a_{2n} \\ \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots a_{mn} \end{bmatrix}$$

This matrix has m rows and n columns. It is denoted by (a_{ij}) and is read as m by n. The first subscript refers to the number of row and the second to the number of columns to which the element belongs. Thus, a_{13} belongs to the 1st row and 3rd column; in general, a_{ij} refers to the element of jth row and ith column. If the matrix has total number of elements = m. n, arranged in m rows and n column, **it is said to be of the order of m by n**, which is often written as m x n matrix.

'Matrix algebra' basically provides methods for dealing with sets of simultaneous relationships involving many variables.

5.3.2 Types of Matrix

A) Rectangular Matrix

A matrix of the order m by n is also called as 'rectangular matrix'. And if m=n, then the matrix is said to be a '**square matrix**' of order 'n'.

B) Column Matrix

If the matrix consists of only one column as $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, then it is called a '**column matrix**' or '**column vector**'.

C) Null Matrix

If all the elements of a matrix are zero, then it is called a 'null matrix' or 'zero matrix'. The null matrix of order m x n is denoted by $O_{m \times n}$.

$$\text{e.g., } 0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

If the elements of a matrix are not all zero, then it is called a '**non-zero matrix**'

The **negative** of a matrix is $[-a_{ij}]_{m \times n}$ and is denoted by $(-A)$.

$$\text{Thus, if } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$$

$$\text{then } -A = \begin{bmatrix} -a_1 & -b_1 & -c_1 \\ -a_2 & -b_2 & -c_2 \end{bmatrix}$$

The 'transpose of a matrix' A is the matrix obtained from the given matrix Δ by changing its rows into columns and columns into rows. It is denoted by A' or A^T . Hence if A be a matrix of order $(m \times n)$, then A^T is of order $(n \times m)$.

$$\text{Thus, the transpose of } \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \text{ is } \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \\ c_1 & c_2 \end{bmatrix}$$

$$\text{Note that, } [A^T]^T = A$$

If the number of columns be equal its number of rows, then the matrix is called a '**square matrix**'.

Some special forms of square matrices:

D) Diagonal Matrix

If all **off-diagonal** elements of a square matrix are zero, that matrix is called diagonal matrix. Thus

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} = (a_{ij})_{n \times n} \text{ is a diagonal matrix if and only if}$$

$$a_{ij} = 0 \text{ for } i \neq j \text{ and } a_{ij} \neq 0 \text{ for at least one } i = j$$

$$\text{For example, } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \text{ is a diagonal matrix.}$$

E) Scalar Matrix

If the diagonal elements of a diagonal matrix be all equal, then the matrix is called a '**scalar matrix**'.

$$\text{e.g., } \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix} \text{ is a scalar matrix } [a]$$

F) Identity Matrix

If the diagonal elements of a diagonal matrix are all equal to 1, it is defined as a ‘unit matrix’ or ‘identity matrix’. A unit matrix of order ‘n’ is denoted by I_n or by $[I]$ or simply by I . Thus,

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} = (a_{ij})_{n \times n} \text{ is an identity matrix if and only if}$$

$$a_{ij} = 0 \text{ for } i \neq j \text{ and } a_{ij} = 1 \text{ for } i = j$$

Example: $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is an identity matrix of order 3.

Note that $I = I'$

5.3.3 Matrix Relations and Operations

5.3.3.1 Equality of Matrices

Two matrices A and B are said to be ‘*conformable for equality*’ if they are of same order.

Two matrices of same order are said to be ‘*equal*’, if and only if, the corresponding elements of the two are equal.

Example: $\begin{bmatrix} 1^2 & 2^2 \\ 3^2 & 4^2 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 9 & 16 \end{bmatrix}$

If the two matrices are not of the same order, then they are ‘*not conformable for equality*’. In other words, matrices of unequal orders cannot be equal by definition.

5.3.3.2 Addition and Subtraction of Matrices

If the matrices A and B are of the same order ($m \times n$), then they can be added or subtracted.

If $A = [a_{ij}]$ and $B = [b_{ij}]$, then $A+B = [a_{ij} + b_{ij}]$ and $A-B = [a_{ij} - b_{ij}]$

In other words, corresponding elements are added or subtracted. The new matrix will be of the same order ($m \times n$).

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+0 & 2+1 & 3+2 \\ 3+2 & 4+3 & 5+4 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 5 & 7 & 9 \end{bmatrix}$

Some Properties:

- i) $A + B = B + A$ (so, matrix addition is commutative)
- ii) $(A + B) + C = A + (B + C)$ where C is the third matrix (so, matrix addition is associative).